



University of
East London

Ain Shams University

Faculty of Computer & Information Sciences
Software Engineering Program | CBSD Course

PathFinder

“Student Management System”

Personalized Career and Learning Roadmaps for CS Students

CBSD Flutter Project Documentation

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1. Introduction

PathFinder "Student Management System" is a cross-platform mobile application developed using Flutter, designed to assist undergraduate Computer Science students at Ain Shams University in identifying the most suitable career path based on their technical and soft skills. The application combines artificial intelligence, cloud services, and native mobile features to deliver a comprehensive career guidance experience.

The application integrates with a Python-based AI backend powered by FastAPI, machine learning models, and Natural Language Processing (NLP) to predict suitable career tracks and provide personalized learning roadmaps. PathFinder supports six main career fields relevant to Computer Science students:

- Artificial Intelligence
- Data Science
- Software Development
- Cybersecurity
- Software Engineering
- UI/UX Design

The Flutter application serves as the mobile frontend that connects students with the AI backend, Firebase cloud services, and external learning resources. This document provides a comprehensive overview of the PathFinder Flutter application including its architecture, design patterns, features, and team contributions.

2. Problem Statement

Computer Science students at Egyptian universities face significant challenges when making career decisions during their academic journey. The core problems identified are:

- **Career Uncertainty:** Students choose career paths without adequate insight into required skills, role alignment, or necessary academic preparation, leading to confusion and wasted effort.
- **Misaligned Choices:** Poor career decisions that do not match the student's actual capabilities, interests, or available opportunities in the job market.
- **Skill-Market Gap:** Significant gaps between the skills acquired during academic studies and the skills demanded by industry, resulting in low employability and job satisfaction.
- **Lack of Personalization:** Existing career guidance platforms are either too generic, not focused on Computer Science domains, or require paid subscriptions.
- **Fragmented Learning Resources:** Available courses and learning materials are scattered across multiple platforms without a unified, personalized, and actionable roadmap.

PathFinder addresses these challenges through an intelligent, AI-driven mobile application that assesses students' current skills, predicts the most suitable career track, and generates a step-by-step personalized learning roadmap with curated course recommendations from Coursera, edX, Kaggle, and other platforms.

3. Objectives

The primary objectives of the PathFinder "Student Management System" Flutter application are:

- Develop a cross-platform Flutter mobile application with a modern, professional UI/UX following Material Design 3 guidelines.
- Implement secure and complete user authentication using Firebase Authentication including registration, login, email verification, and password reset.
- Integrate Firebase Firestore for persistent cloud storage of quiz results and user data across devices.
- Build an AI-powered skill assessment quiz that evaluates students across five key domains: data analysis, coding, cybersecurity, UI/UX design, and artificial intelligence.
- Display personalized career predictions with skill breakdown visualizations and generate downloadable PDF career reports.
- Provide an interactive AI Chat interface connected to the Python backend for real-time career guidance and advice.
- Implement native device features including camera access for profile management and system share functionality.
- Integrate URL Launcher to direct students to relevant free online courses matching their career track.
- Create a reusable CareerBadgeWidget custom component published to a separate GitHub repository.
- Utilize the step_progress_indicator third-party package for professional roadmap progress visualization.
- Deliver bonus features including PDF report generation and email notifications via Firebase.

4. Project Design

4.1 System Architecture

PathFinder "Student Management System" follows a clean three-layer architecture that separates concerns between presentation, application logic, and data management:

Layer	Component	Technology	Responsibility
Presentation	Login / Register Screen	Flutter + Firebase Auth	User authentication UI
Presentation	Home Screen	Flutter + NavigationBar	Main dashboard with career tracks
Presentation	Quiz Screen	Flutter StatefulWidget	5-question skill assessment
Presentation	Result Screen	Flutter + Share Plus	Career prediction display
Presentation	Roadmap Screen	Flutter + StepProgressIndicator	Learning path visualization
Presentation	Chat Screen	Flutter StatefulWidget	AI chat interface
Presentation	Profile Screen	Flutter + ImagePicker	User profile management
Application	ApiService	Dart + HTTP	Backend API communication
Application	NotificationService	Flutter SnackBar	In-app notifications
Application	PdfService	pdf + printing packages	PDF report generation
Application	CareerBadgeWidget	Flutter Custom Widget	Reusable career card component
Data	Firebase Auth	Firebase Authentication	User registration & login
Data	Firebase Firestore	Cloud Firestore	Quiz results cloud storage
Data	SharedPreferences	Local Storage	User data local persistence
Data	Python Backend	FastAPI + ML Models	AI career prediction & chat

4.2 Application Screens (8 Screens)

The application consists of 8 fully functional screens connected through Bottom Navigation and Stack navigation:

#	Screen	Widget Type	Key Features
1	Login Screen	StatefulWidget	Firestore Auth, Forgot Password, SVG Logo
2	Register Screen	StatefulWidget	Firestore Auth, Email Verification, University & Program fields
3	Home Screen	StatefulWidget	Career Tracks grid with illustrations, API status indicator
4	Quiz Screen	StatefulWidget	5 animated questions, progress indicator per domain
5	Result Screen	StatefulWidget	Career prediction, Save to Firestore, Share, PDF download
6	Roadmap Screen	StatelessWidget	StepProgressIndicator, course links via URL Launcher
7	AI Chat Screen	StatefulWidget	Real-time chat with Python AI backend
8	Profile Screen	StatefulWidget	Image Picker (Camera), Firestore data, Logout

4.3 Use Case Diagram

The following table represents the use cases for the PathFinder system, describing all interactions between the Student actor and the system:

Actor	Use Case	Description
Student	Register Account	Create new account with name, email, university, program
Student	Login	Sign in using Firebase Authentication
Student	Reset Password	Receive password reset email via Firebase
Student	Take Skill Quiz	Answer 5 questions to assess skills
Student	View Career Result	See recommended career track after quiz
Student	Save Result	Save result to Firestore and SharedPreferences
Student	Share Result	Share career result via Share Plus
Student	Download PDF	Generate and download PDF career report
Student	View Roadmap	View personalized learning roadmap with steps
Student	Open Course Link	Launch course URL via URL Launcher
Student	Chat with AI	Send messages to AI career assistant
Student	Update Profile Picture	Select photo from gallery using Image Picker
Student	Logout	Clear session and return to login screen
System	Send Verification Email	Firebase sends email verification on register
System	Store Data (Firestore)	Save quiz results to Firebase Firestore
System	Store Data (Local)	Save user data to SharedPreferences

4.4 Class Diagram

The following table describes the main classes, their types, attributes, and methods in the PathFinder Flutter application:

Class Name	Type	Attributes	Methods
LoginScreen	StatefulWidget	emailController, passwordController, isLoading	login(), forgotPassword()
RegisterScreen	StatefulWidget	nameController, emailController, passwordController, universityController, programController	register()
HomeScreen	StatefulWidget	currentIndex, screens[]	build()
QuizScreen	StatefulWidget	currentQuestion, answers{}, questions[]	answer(value)
ResultScreen	StatefulWidget	saved, isSaving, answers{}	getCareerTrack(), saveResult(), shareResult(), downloadPdf()
RoadmapScreen	StatelessWidget	careerTrack	getRoadmap(), openUrl(), getTrackColor()
ChatScreen	StatefulWidget	messages[], isLoading, messageController	sendMessage(), scrollToBottom()
ProfileScreen	StatefulWidget	name, email, imagePath, careerTrack, university, program	loadData(), pickImage(), logout()
CareerBadgeWidget	StatefulWidget	title, icon, color, isSelected, imagePath, onTap	build() with animation
ApiService	class (static)	baseUrl	isOnline(), saveQuizResult(), getSavedCareerTrack(), sendMessage()
NotificationService	class (static)	-	init(), showNotification()
PdfService	class (static)	-	generateCareerReport()
AppTheme	class (static)	primary, secondary, background, surface, card, textPrimary, textSecondary	lightTheme

4.5 Navigation Flow

PathFinder uses two navigation patterns:

- Bottom Navigation Bar (Tabs): Main navigation between Home, Quiz, AI Chat, Roadmap, and Profile screens.
- Stack Navigation: Push/pop navigation for transitioning between Login → Register, Quiz → Result → Roadmap screens.

4.6 Stateful & Stateless Widget Combination

The application demonstrates a clear combination of both widget types as required:

- Stateful Widgets: LoginScreen, RegisterScreen, HomeScreen, QuizScreen, ResultScreen, ProfileScreen, ChatScreen, CareerBadgeWidget — these manage changing state such as form input, quiz progress, API responses, and image selection.
- Stateless Widgets: RoadmapScreen, _InfoTile, _MessageBubble, _SuggestionChip, _ActionButton — these display static content derived from passed parameters.

4.7 Components & Packages Used

Package	Version	Purpose
firebase_core	^3.6.0	Firebase initialization
firebase_auth	^5.3.1	User authentication & email verification
cloud_firestore	^5.4.4	Cloud database for quiz results
shared_preferences	^2.2.2	Local storage for user data
http	^1.2.0	REST API integration with Python backend
image_picker	^1.0.7	Camera/Gallery access (Native Feature)
share_plus	^13.1.0	Share career results (Native Feature)
url_launcher	^6.3.0	Open course URLs in browser
step_progress_indicator	^1.0.2	Custom progress UI (3rd party component)
flutter_svg	^2.0.9	SVG image rendering
google_fonts	^6.1.0	Poppins font integration
pdf + printing	^3.10.8	PDF career report generation (Bonus)
path_provider	^2.1.2	File system access for PDF

5. Who Did What

The following table details the contribution and responsibilities of each team member throughout the project:

Team Member	Tasks & Responsibilities
Khaled Sami Gad	Project management, Firebase Authentication implementation (Login, Register, Email Verification, Forgot Password), Firebase Firestore integration, Google Services configuration, GitHub repository management, Project documentation.
Ganna Mohammed Safar	Flutter project setup & architecture, Home Screen design, CareerBadgeWidget custom component (GitHub upload), Bottom Navigation, API integration with Python backend, Notification service, App theme & color system.
Mohamed Salah Sayed	Quiz Screen with animated options, Result Screen, PDF report generation (Bonus), Share functionality (Bonus), Career prediction logic, Skills breakdown visualization, URL Launcher integration for courses Firestore data saving.
Mahmoud Abdelmenam	Roadmap Screen with StepProgressIndicator (3rd party component), Firestore data saving, Chat Screen with AI backend, Profile Screen with camera feature, Notification service, SVG illustrations integration.

6. Conclusion

PathFinder”Student Management System” successfully delivers a comprehensive, production-quality Flutter mobile application that meets and exceeds all CBSD course requirements. The application provides real value to Computer Science students by combining AI-powered career prediction with personalized learning roadmaps.

All mandatory requirements have been fulfilled:

- Authentication: Full Firebase Authentication with email/password login, registration, email verification, and forgot password functionality.
- Plugin: image_picker plugin integrated for native camera/gallery access in the Profile screen.
- Native Features: Camera access via ImagePicker, system Share functionality via share_plus, and URL Launcher for opening course links.
- Navigation: Bottom Navigation Bar (Tabs) combined with Stack navigation throughout the app.
- Stateful & Stateless Widgets: Clear combination across all 8 screens and shared components.
- Assets: Poppins font family (Regular, Bold, SemiBold) and SVG career track illustrations from unDraw.
- Local Storage: SharedPreferences stores user name, email, university, program, career track, and quiz answers.
- API Integration: REST API integration with Python FastAPI backend for AI career prediction and chat.

Bonus features delivered include PDF career report generation with download functionality, email verification on registration, forgot password via Firebase, Firebase Firestore cloud database for quiz results, and an AI Chat screen connected to the Python backend.

The CareerBadgeWidget custom component was developed and published to a separate GitHub repository (github.com/Khaled-26/career-badge-widget), demonstrating reusable component design principles.

7. References

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